NEET UG 2027: Master Chapter Theory

Physical Chemistry: In-Depth Derivations & High-Yield Archetypes

Prepared, Compiled & Authored by Satya Prakash · Strictly Calibrated to Revised NMC Syllabus

1 NMC 2027 Regulatory Calibration & Scope Demarcation

Zero-Tolerance NMC Syllabus Alert: Deleted Physical Chemistry Chapters

Under the revised National Medical Commission (NMC) NEET UG 2027 regulatory curriculum, the following legacy chapters have been **strictly deleted and removed from the examination**:

1. **Solid State:** All unit cells, crystal systems, Bragg's diffraction, packing efficiency derivations, point defects (Schottky and Frenkel defects), electrical and magnetic properties of solids are **completely eliminated**.
2. **States of Matter (Gaseous & Liquid Bulk States):** Macroscopic gas laws, van der Waals equation of state, critical phenomena (P_c, V_c, T_c), compressibility factor (Z) curves, liquefaction of gases, and bulk liquid properties (surface tension, viscosity) are **completely eliminated**. (Note: Ideal gas stoichiometry at STP is retained in Unit 1; molecular kinetic theory is covered in Physics).
3. **Surface Chemistry:** Adsorption isotherms (Freundlich and Langmuir), colloidal states, lyophilic and lyophobic sols, micelles and CMC, Tyndall effect, electrophoresis, coagulation and Hardy-Schulze rule, and heterogeneous catalysis are **strictly eliminated** from theory.

Study Directive: Do not invest study hours in these eliminated topics. All cognitive focus must remain on the rigorously tested Class 11 and Class 12 physical chemistry units detailed below.

2 Some Basic Concepts in Chemistry: Mole Concept & Stoichiometry (Unit 1)

2.1 Laws of Chemical Combination

Chemical transformations obey five fundamental empirical laws that form the bedrock of quantitative chemistry:

1. **Law of Conservation of Mass (Lavoisier, 1789):** In any chemical reaction, the total mass of the reactants equals the total mass of the products: $\sum m_{\text{reactants}} = \sum m_{\text{products}}$.
2. **Law of Definite Proportions (Proust, 1799):** A given chemical compound always contains exactly the same proportion of elements by mass, regardless of its source or method of preparation.
3. **Law of Multiple Proportions (Dalton, 1803):** If two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in the ratio of small whole numbers (e.g., in CO and CO_2, fixed 12 g of carbon combines with 16 g and 32 g of oxygen, ratio 1 : 2).
4. **Gay-Lussac's Law of Gaseous Volumes (1808):** When gases react together, they do so in volumes which bear a simple whole-number ratio to one another and to the volumes of gaseous products, provided all volumes are measured at identical temperature and pressure.
5. **Avogadro's Hypothesis (1811):** Equal volumes of all gases under identical conditions of temperature and pressure contain an equal number of molecules: $V \propto n$ (at constant T, P).

2.2 The Mole Concept & Molar Quantities

The mole is the SI base unit for the amount of substance. Exactly one mole contains $6.02214076 \times 10^{23}$ elementary entities (Avogadro's constant, N_A).

Master Mole Formulations

For any pure chemical substance of mass w (in g), molar mass M (in $\mathrm{g\,mol^{-1}}$), and elementary particle count N:

$$n = \frac{w}{M} = \frac{N}{N_A} = \frac{V_{\mathrm{STP}}\ (\text{in Litres})}{V_m} \tag{1}$$

where the standard molar volume V_m depends on the adopted thermodynamic convention:

- **Traditional NCERT Standard** ($T = 273.15\,\mathrm{K}, P = 1\,\mathrm{atm} = 101.325\,\mathrm{kPa}$)**:** $V_m = 22.414\,\mathrm{L\,mol^{-1}} \approx 22.4\,\mathrm{L}$.
- **IUPAC Standard** ($T = 273.15\,\mathrm{K}, P = 1\,\mathrm{bar} = 100.0\,\mathrm{kPa}$)**:** $V_m = 22.711\,\mathrm{L\,mol^{-1}} \approx 22.7\,\mathrm{L}$.

Vapour Density (VD) Relationship for Ideal Gases:

$$\mathrm{VD} = \frac{\rho_{\mathrm{gas}}}{\rho_{\mathrm{H_2}}} = \frac{M_{\mathrm{gas}}}{M_{\mathrm{H_2}}} = \frac{M_{\mathrm{gas}}}{2.016} \implies M \approx 2 \times \mathrm{VD} \tag{2}$$

2.3 Empirical and Molecular Formulas

Empirical vs Molecular Formula Determination

- **Empirical Formula (EF):** The simplest whole-number ratio of atoms of each element present in a compound.
- **Molecular Formula (MF):** The actual number of atoms of each element present in a molecule of the compound:

$$\text{Molecular Formula} = (\text{Empirical Formula})_n, \qquad n = \frac{\text{Molar Mass}}{\text{Empirical Formula Mass}} \tag{3}$$

- **Algorithmic Determination:**
  1. Convert given mass percentages directly to mass in 100 g of sample.
  2. Divide each mass by the respective atomic mass to compute relative moles: $n_i = w_i/A_i$.
  3. Divide each n_i by the smallest mole value among them: $r_i = n_i/n_{\min}$.
  4. If the resulting ratios are non-integers, multiply all ratios by the smallest integer that converts them into whole numbers.

2.4 Stoichiometry, Limiting Reagents & Reaction Yields

For a general balanced stoichiometric chemical equation:

$$aA + bB \longrightarrow cC + dD \tag{4}$$

The stoichiometric coefficients a, b, c, d represent the exact molar combining ratios:

$$\frac{n_A\ (\text{consumed})}{a} = \frac{n_B\ (\text{consumed})}{b} = \frac{n_C\ (\text{produced})}{c} = \frac{n_D\ (\text{produced})}{d} \tag{5}$$

Limiting Reagent (LR) Diagnostic Algorithm

When arbitrary initial quantities $n_{A,0}$ and $n_{B,0}$ are supplied:

1. Compute the mole-to-coefficient ratio for each reactant:

$$\mathcal{R}_A = \frac{n_{A,0}}{a}, \qquad \mathcal{R}_B = \frac{n_{B,0}}{b} \tag{6}$$

2. If $\mathcal{R}_A < \mathcal{R}_B$, reactant A is the **Limiting Reagent** (completely consumed). Reactant B is in excess.
3. All product yields and reactant consumptions are strictly determined by the limiting reagent:

$$n_C = c\left(\frac{n_{A,0}}{a}\right), \qquad n_{B,\text{remaining}} = n_{B,0} - b\left(\frac{n_{A,0}}{a}\right) \tag{7}$$

4. **Percentage Yield & Percentage Purity:**

$$\text{Percentage Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100\% \tag{8}$$

$$\text{Percentage Purity} = \frac{\text{Mass of pure reactant}}{\text{Total mass of impure sample}} \times 100\% \tag{9}$$

2.5 Solution Concentration Expressions & Interconversions

The quantitative composition of solutions is formulated across six distinct mathematical regimes:

Concentration Unit	Symbol	Mathematical Definition	Temperature Dependence
Molarity	M	$M = \frac{n_{\text{solute}}}{V_{\text{sol}}\ (\text{in L})} = \frac{w_B \times 1000}{M_B \times V_{\text{sol}}\ (\text{in mL})}$	**Temperature Dependent** ($V \propto T \implies M$ decreases as T rises)
Molality	m	$m = \frac{n_{\text{solute}}}{W_{\text{solvent}}\ (\text{in kg})} = \frac{w_B \times 1000}{M_B \times W_A\ (\text{in g})}$	**Temperature Independent** (mass is invariant with temperature)
Normality	N	$N = \frac{\text{Gram Equivalents}}{V_{\text{sol}}\ (\text{in L})} = M \times n\text{-factor}$	**Temperature Dependent** ($V \propto T$)
Mole Fraction	x_B	$x_B = \frac{n_B}{n_A + n_B}$, $x_A + x_B = 1$	**Temperature Independent**
Mass Percentage	$\%(w/w)$	$\%(w/w) = \frac{w_{\text{solute}}}{w_{\text{solution}}} \times 100$	**Temperature Independent**
Volume Percentage	$\%(v/v)$	$\%(v/v) = \frac{V_{\text{solute}}}{V_{\text{solution}}} \times 100$	**Temperature Dependent**
Parts Per Million	ppm	$\text{ppm} = \frac{w_{\text{solute}}}{w_{\text{solution}}} \times 10^6$	**Temperature Independent**

Table 1: Comprehensive Classification of Solution Concentration Regimes.

Master Interconversion Formulations (d = density in g mL^{-1})

$$M = \frac{10 \times \%(w/w) \times d}{M_B} \tag{10}$$

$$m = \frac{1000 \times M}{1000\,d - M\,M_B} \tag{11}$$

$$m = \frac{x_B \times 1000}{(1 - x_B)M_A} = \frac{x_B \times 1000}{x_A M_A} \tag{12}$$

$$x_B = \frac{m\,M_A}{1000 + m\,M_A} \tag{13}$$

Dilution and Mixing Formulations:

$$M_1V_1 = M_2V_2, \qquad M_{\text{mix}} = \frac{\sum M_iV_i}{\sum V_i}, \qquad N_1V_1 = N_2V_2 \quad \text{(at exact redox/acid-base equivalence)} \tag{14}$$

NCERT Benchmark: Valence Factor (n-factor) Rules

- **Acids:** n-factor = basicity (number of replaceable H^+ ions): HCl $\implies$ 1; H_2SO_4 $\implies$ 2; H_3PO_4 $\implies$ 3; H_3PO_3 $\implies$ 2 (one P – H bond is non-ionizable!); H_3PO_2 $\implies$ 1 (two P – H bonds are non-ionizable!).
- **Bases:** n-factor = acidity (number of replaceable OH^- ions): NaOH $\implies$ 1; $Ca(OH)_2$ $\implies$ 2; $Al(OH)_3$ $\implies$ 3.
- **Salts:** n-factor = total positive or negative charge: NaCl $\implies$ 1; $CaCl_2$ $\implies$ 2; $Al_2(SO_4)_3$ $\implies$ 6.
- **Redox Agents:** n-factor = total change in oxidation state per mole of reagent.

Common Pitfall: STP Molar Volume & Equivalent Mass of Oxyacids

- **STP Standard Conditions Shift:** In modern IUPAC / NCERT convention, standard temperature and pressure (STP) is defined at $T = 273.15\,\text{K}$ and $P = 1\,\text{bar} = 10^5\,\text{Pa}$. Under these modern conditions, the molar volume of an ideal gas is $V_\text{m} = 22.7\,\text{L}\cdot\text{mol}^{-1}$ (or $22.7\,\text{dm}^3\cdot\text{mol}^{-1}$). The legacy convention ($1\,\text{atm} = 1.01325\times10^5\,\text{Pa}$) yielded $22.4\,\text{L}\cdot\text{mol}^{-1}$. Always inspect whether numerical question options in NEET assume legacy (22.4 L) or modern IUPAC (22.7 L).
- **Polyprotic Phosphorus Oxyacids:** Never equate molecular mass divided by 3 to equivalent mass blindly! For orthophosphoric acid (H_3PO_4), basicity is 3 $\implies E = M/3$. For phosphorous acid (H_3PO_3), basicity is 2 $\implies E = M/2$ because one hydrogen is directly bonded to phosphorus (P – H) and does not ionize in aqueous solution. For hypophosphorous acid (H_3PO_2), basicity is 1 $\implies E = M/1$ since two hydrogens are non-ionizable P – H bonds.

3 Atomic Structure & Wave Mechanics (Unit 2)

3.1 Dual Nature of Light & The Photoelectric Effect

Einstein formulated the photoelectric effect by treating light as discrete localized wave packets (photons) of energy $E = h\nu$. When a photon strikes a metallic surface:

$$E_\text{photon} = \Phi + K_\text{max} \implies h\nu = h\nu_0 + \frac{1}{2}m_e v_\text{max}^2 = h\nu_0 + eV_0 \tag{15}$$

where $\Phi = h\nu_0$ is the work function of the metal, ν_0 is the threshold frequency, and V_0 is the stopping potential.

3.2 Bohr's Model of the Hydrogen Atom: Complete Mathematical Derivations

Niels Bohr synthesized classical electrostatic planetary mechanics with Planckian angular momentum quantization for hydrogen-like species of nuclear charge $+Ze$ and single orbital electron $-e$:

1. **Postulate 1 (Electrostatic Centripetal Equilibrium):**

$$\frac{m_e v_n^2}{r_n} = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n^2} \implies m_e v_n^2 r_n = \frac{Ze^2}{4\pi\varepsilon_0} \tag{16}$$

2. **Postulate 2 (Quantization of Orbital Angular Momentum):**

$$L_n = m_e v_n r_n = \frac{nh}{2\pi} \quad (n = 1, 2, 3, \dots) \tag{17}$$

3.2.1 Derivation of Quantized Orbit Radius (r_n)

Dividing Equation 17 by $m_e r_n$ yields the electron orbital velocity:

$$v_n = \frac{nh}{2\pi m_e r_n} \tag{18}$$

Substituting this expression for v_n into Equation 16:

$$m_e\left(\frac{nh}{2\pi m_e r_n}\right)^2 r_n = \frac{Ze^2}{4\pi\varepsilon_0} \implies \frac{n^2h^2}{4\pi^2 m_e r_n} = \frac{Ze^2}{4\pi\varepsilon_0} \tag{19}$$

Solving explicitly for r_n:

$$r_n = \frac{\varepsilon_0 h^2}{\pi m_e e^2}\frac{n^2}{Z} = a_0\frac{n^2}{Z} \approx 0.529\,\text{Å} \times \frac{n^2}{Z} = 52.9\,\text{pm} \times \frac{n^2}{Z} \tag{20}$$

where $a_0 = 0.529177\,\text{Å}$ is the first Bohr radius of hydrogen ($n = 1, Z = 1$).

3.2.2 Derivation of Quantized Velocity (v_n)

Substituting r_n into Equation 17:

$$v_n = \frac{nh}{2\pi m_e \left(\frac{\varepsilon_0 h^2 n^2}{\pi m_e e^2 Z}\right)} = \frac{e^2}{2\varepsilon_0 h}\frac{Z}{n} = v_0\frac{Z}{n} \approx 2.188 \times 10^6 \frac{Z}{n}\,\mathrm{m\,s^{-1}} = \frac{c}{137}\frac{Z}{n} \tag{21}$$

3.2.3 Derivation of Quantized Energies (K_n, U_n, E_n)

$$K_n = \frac{1}{2}m_e v_n^2 = \frac{1}{2}\left(\frac{Ze^2}{4\pi\varepsilon_0 r_n}\right) = \frac{Ze^2}{8\pi\varepsilon_0 r_n} \tag{22}$$

$$U_n = -\frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r_n} = -2K_n \quad \text{(Virial Theorem: } 2\langle K\rangle + \langle U\rangle = 0) \tag{23}$$

$$E_n = K_n + U_n = -K_n = \frac{1}{2}U_n = -\frac{Ze^2}{8\pi\varepsilon_0 r_n} \tag{24}$$

Substituting the derived value of r_n:

$$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2}\frac{Z^2}{n^2} = -R_H hc\frac{Z^2}{n^2} = -2.18 \times 10^{-18}\,\mathrm{J} \times \frac{Z^2}{n^2} = -13.6\frac{Z^2}{n^2}\,\mathrm{eV} \tag{25}$$

3.2.4 Spectral Transitions & The Rydberg Equation

When an electron drops from an initial upper energy level n_2 to a lower level n_1 ($n_2 > n_1$):

$$\Delta E = E_{n_2} - E_{n_1} = h\nu = \frac{hc}{\lambda} \implies \bar{\nu} = \frac{1}{\lambda} = R_H Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \tag{26}$$

where $R_H = \frac{m_e e^4}{8\varepsilon_0^2 h^3 c} \approx 1.09677 \times 10^7\,\mathrm{m^{-1}}$ ($1/R_H \approx 912\,\text{Å}$).

Spectral Series	**Lower Level (n_1)**	**Upper Level (n_2)**	**Spectral Region**	**Wavelength Range for H ($Z = 1$)**
Lyman	1	$2, 3, 4, \ldots, \infty$	Ultraviolet (UV)	$912\,\text{Å} \leq \lambda \leq 1216\,\text{Å}$
Balmer	2	$3, 4, 5, \ldots, \infty$	Visible (H_α to H_δ) + Near UV	$3646\,\text{Å} \leq \lambda \leq 6563\,\text{Å}$
Paschen	3	$4, 5, 6, \ldots, \infty$	Near Infrared (IR)	$8204\,\text{Å} \leq \lambda \leq 18751\,\text{Å}$
Brackett	4	$5, 6, 7, \ldots, \infty$	Infrared (IR)	$14580\,\text{Å} \leq \lambda \leq 40512\,\text{Å}$
Pfund	5	$6, 7, 8, \ldots, \infty$	Far Infrared (Far IR)	$22790\,\text{Å} \leq \lambda \leq 74580\,\text{Å}$

Table 2: Hydrogen Atom Emission Spectral Series.

3.3 Wave-Particle Duality: The de Broglie Relation

Louis de Broglie (1924) postulated that matter exhibits intrinsic wave characteristics. Equating Einstein's mass-energy equivalence ($E = mc^2 = pc$) with Planck's photon energy ($E = h\nu = hc/\lambda$):

$$p = \frac{h}{\lambda} \implies \lambda = \frac{h}{p} = \frac{h}{mv} \tag{27}$$

For a charged particle of mass m and charge q accelerated from rest through an electrostatic potential difference V:

$$K = qV = \frac{p^2}{2m} \implies p = \sqrt{2mqV} \implies \lambda = \frac{h}{\sqrt{2mqV}} \tag{28}$$

Specialized de Broglie Wavelength Formulas:

- **Accelerated Electron ($m = m_e, q = e$):** $\lambda_e = \frac{1.227}{\sqrt{V}}\,\text{nm} = \frac{12.27}{\sqrt{V}}\,\text{Å}$.
- **Thermal Neutron / Gas Molecule ($K = \frac{3}{2}k_BT$):** $\lambda_{\text{thermal}} = \frac{h}{\sqrt{3mk_BT}}$.

3.4 Heisenberg's Uncertainty Principle

Werner Heisenberg (1927) proved that it is physically impossible to simultaneously determine both the exact position and exact momentum of a microscopic quantum entity:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2} = \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v_x) \geq \frac{h}{4\pi} \implies \Delta x \cdot \Delta v_x \geq \frac{h}{4\pi m} \tag{29}$$

Alternative Conjugate Variables Formulations:

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}, \qquad \Delta\theta \cdot \Delta L \geq \frac{h}{4\pi} \tag{30}$$

Because Planck's constant $h = 6.626 \times 10^{-34}\,\text{J s}$ is infinitesimally small, the uncertainty product is completely imperceptible for macroscopic bodies ($m \sim 1\,\text{kg}$), but dominates the atomic regime ($m_e \sim 9.1 \times 10^{-31}\,\text{kg}$).

3.5 Schrödinger Wave Equation & Quantum Mechanical Formalism

Erwin Schrödinger (1926) formulated the stationary-state wave equation for a quantum particle of mass m moving in a potential field $V(x, y, z)$:

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \iff \hat{H}\psi = E\psi \tag{31}$$

where $\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V$ is the Hamiltonian operator, and $\psi(r, \theta, \phi)$ is the eigenfunction.

Physical Interpretation of Wave Function ψ and Probability Density $|\psi|^2$

- **Wave Function Amplitude (ψ):** ψ possesses no direct physical meaning; it can be positive, negative, or complex.
- **Born's Probability Density ($|\psi|^2 = \psi^*\psi$):** Represents the probability of finding the electron per unit volume in an infinitesimal volume element $d\tau = dx\,dy\,dz$:

$$dP = |\psi|^2 d\tau, \qquad \int_{\text{all space}} |\psi|^2 d\tau = 1 \quad \text{(Normalization Condition)} \tag{32}$$

- **Radial vs Angular Separation:** In spherical coordinates (r, θ, ϕ):

$$\psi(r, \theta, \phi) = R_{n,l}(r) \times Y_{l,m_l}(\theta, \phi) \tag{33}$$

  $R_{n,l}(r)$ (radial function) dictates orbital size and radial probability; $Y_{l,m_l}(\theta, \phi)$ (angular function) dictates orbital 3D spatial geometry and orientation.
- **Radial Probability Distribution Function ($4\pi r^2 R^2(r)$):** Represents the absolute probability of locating the electron in a spherical shell of radius r and thickness dr:

$$P(r)dr = 4\pi r^2 R^2(r) dr \tag{34}$$

  At $r = 0$, $4\pi r^2 R^2 = 0$ for all orbitals (electron cannot reside at the mathematical point of the nucleus).

Master Nodal Formulations

Nodes are spatial domains where the probability density $|\psi|^2$ vanishes identically:

$$\text{Radial Nodes (Spherical Nodes)} = n - l - 1 \tag{35}$$

$$\text{Angular Nodes (Nodal Planes / Cones)} = l \tag{36}$$

$$\text{Total Nodes} = (n - l - 1) + l = n - 1 \tag{37}$$

Examples:

- $1s$: Radial $= 1 - 0 - 1 = 0$, Angular $= 0$, Total $= 0$.
- $2s$: Radial $= 2 - 0 - 1 = 1$, Angular $= 0$, Total $= 1$.

- $2p$: Radial $= 2-1-1=0$, Angular $=1$ (the planar surface passing through nucleus), Total $=1$.
- $3d$: Radial $= 3-2-1=0$, Angular $=2$ (two nodal planes/cones), Total $=2$.
- $4d$: Radial $= 4-2-1=1$, Angular $=2$, Total $=3$.

3.6 Quantum Numbers & Rules of Electron Configuration

An atomic electron is completely described by four quantum numbers:

1. **Principal Quantum Number ($n \in \{1,2,3,\dots\}$):** Determines main energy shell, average distance from nucleus, maximum electrons $= 2n^2$, maximum orbitals $= n^2$.
2. **Azimuthal Quantum Number ($l \in \{0,1,2,\dots,n-1\}$):** Determines subshell shape ($s:0, p:1, d:2, f:3$), and orbital angular momentum:

$$L_{\text{orbital}} = \sqrt{l(l+1)}\hbar = \sqrt{l(l+1)}\frac{h}{2\pi} \quad (\text{For any } s\text{-orbital}, l=0 \implies L=0!) \tag{38}$$

3. **Magnetic Quantum Number ($m_l \in \{-l,\dots,0,\dots,+l\}$):** Determines 3D spatial orientation of the orbital in an external magnetic field ($2l+1$ degenerate orientations).
4. **Spin Quantum Number ($m_s \in \{+\frac{1}{2}, -\frac{1}{2}\}$):** Intrinsic angular momentum orientation. Spin-only magnetic moment:

$$\mu_s = \sqrt{n_u(n_u+2)}\,\text{BM} \quad (n_u = \text{number of unpaired electrons, BM} = \text{Bohr Magneton}) \tag{39}$$

Fundamental Rules of Electronic Architecture

- **Aufbau Principle:** In the ground state of an atom, electrons occupy the lowest available atomic energy orbitals first. Energy follows the $(n+l)$ **rule**:
  1. Lower $(n+l)$ orbital fills before higher $(n+l)$ (e.g., $4s: 4+0=4$ fills before $3d: 3+2=5$).
  2. If two orbitals share identical $(n+l)$, the orbital with lower n fills first (e.g., $3d: 3+2=5$ fills before $4p: 4+1=5$).
- **Pauli's Exclusion Principle:** No two electrons in an atom can have the identical set of all four quantum numbers. Consequently, an individual orbital accommodates at most two electrons with antiparallel spins ($\uparrow\downarrow$).
- **Hund's Rule of Maximum Multiplicity:** Pairing of electrons in degenerate orbitals (p, d, f) cannot occur until each orbital of the subshell is singly occupied with parallel spins.
- **Extra Stability of Half-Filled and Fully-Filled Subshells:**
  - **Symmetrical Electron Distribution:** Spherically symmetric charge minimizes interelectronic repulsions.
  - **Exchange Energy:** Quantum mechanical exchange of electrons with parallel spins releases stabilization energy. For n electrons of parallel spin:

$$\text{Number of possible exchanges} = \frac{n(n-1)}{2} \tag{40}$$

For Chromium ($[\text{Ar}]3d^5 4s^1$): $3d^5$ allows $\frac{5\times4}{2}=10$ exchanges, vs $[\text{Ar}]3d^4 4s^2$ which allows only $\frac{4\times3}{2}=6$ exchanges. Copper ($[\text{Ar}]3d^{10}4s^1$) similarly gains maximum exchange stability over $3d^9 4s^2$.

4 Chemical Thermodynamics & Thermochemistry (Unit 4)

4.1 Fundamental Concepts: Systems, Variables & Functions

- **Thermodynamic System:** That part of the universe selected for thermodynamic study.
  - **Open System:** Exchanges both matter and energy with surroundings (e.g., open beaker of boiling water).
  - **Closed System:** Exchanges energy but not matter with surroundings (e.g., sealed metal canister).
  - **Isolated System:** Exchanges neither matter nor energy with surroundings (e.g., perfectly insulated Dewar flask).

- **State Functions vs Path Functions:**
  - **State Functions:** Properties whose values depend solely upon the current thermodynamic state of the system, independent of the path traversed (U, H, S, G, P, V, T). Cyclic integral $\oint dX = 0$.
  - **Path Functions:** Quantities whose numerical values depend explicitly upon the transition path traversed between states ($q = \text{heat}, w = \text{work}$). Cyclic integral $\oint dq \neq 0$.
- **Intensive vs Extensive Properties:**
  - **Extensive (additive, depends on mass):** Mass (m), volume (V), internal energy (U), enthalpy (H), entropy (S), Gibbs energy (G), heat capacity (C).
  - **Intensive (scale-independent, independent of mass):** Temperature (T), pressure (P), density (ρ), molar heat capacity (C_m), specific heat (c), viscosity (η), surface tension, refractive index, EMF of a cell (E_{cell}).

4.2 First Law of Thermodynamics & Expansion Work Derivations

The First Law is the conservation of energy principle applied to thermodynamic systems:

$$\Delta U = q + w \tag{41}$$

where $\Delta U = U_2 - U_1$ is the change in internal energy, q is the heat absorbed by the system ($q > 0$ if endothermic, $q < 0$ if exothermic), and w is work done on the system ($w > 0$ for compression, $w < 0$ for expansion under IUPAC sign convention).

4.2.1 Derivation of Reversible Isothermal Expansion Work of an Ideal Gas

For an infinitesimal quasi-static reversible expansion, the internal gas pressure P_{gas} differs from the external opposing pressure P_{ext} by an infinitesimal amount dP ($P_{\text{ext}} = P - dP$):

$$\delta w_{\text{rev}} = -P_{\text{ext}} dV = -(P - dP)dV = -PdV + dPdV \approx -PdV \tag{42}$$

Integrating from initial state (P_1, V_1) to final state (P_2, V_2) at constant temperature T:

$$w_{\text{rev}} = -\int_{V_1}^{V_2} PdV = -\int_{V_1}^{V_2} \frac{nRT}{V} dV = -nRT \ln\left(\frac{V_2}{V_1}\right) = -2.303 nRT \log_{10}\left(\frac{V_2}{V_1}\right) \tag{43}$$

By Boyle's Law ($P_1 V_1 = P_2 V_2$ at constant T):

$$w_{\text{rev}} = -2.303 nRT \log_{10}\left(\frac{P_1}{P_2}\right) \tag{44}$$

Since temperature is constant in an isothermal ideal gas expansion ($\Delta U = 0$):

$$q_{\text{rev}} = -w_{\text{rev}} = +2.303 nRT \log_{10}\left(\frac{V_2}{V_1}\right) \tag{45}$$

4.2.2 Irreversible Expansion & Free Expansion

- **Irreversible Expansion against Constant External Pressure (P_{ext}):**

$$w_{\text{irr}} = -\int_{V_1}^{V_2} P_{\text{ext}} dV = -P_{\text{ext}}(V_2 - V_1) = -P_{\text{ext}}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right) \tag{46}$$

  Because $|w_{\text{rev}}| > |w_{\text{irr}}|$, reversible expansion produces the **maximum work output**.
- **Free Expansion into Vacuum ($P_{\text{ext}} = 0$):**

$$w_{\text{free}} = -\int 0 \cdot dV = 0 \tag{47}$$

  For an ideal gas in an insulated container undergoing free expansion: $w = 0, q = 0 \implies \Delta U = 0 \implies \Delta T = 0$ (Joule's Law).

4.3 Enthalpy, Heat Capacities & Mayer's Relation Derivation

Enthalpy is defined as the total heat content:

$$H = U + PV \implies \Delta H = \Delta U + \Delta(PV) \tag{48}$$

For a chemical reaction involving gases at constant temperature:

$$\Delta H = \Delta U + \Delta n_g RT \tag{49}$$

where $\Delta n_g = \sum n_g(\text{gaseous products}) - \sum n_g(\text{gaseous reactants})$.

Derivation of Mayer's Relation: $C_p - C_v = R$

Heat capacities are defined by:

$$C_v = \left(\frac{\partial U}{\partial T}\right)_V = \frac{q_V}{\Delta T}, \qquad C_p = \left(\frac{\partial H}{\partial T}\right)_P = \frac{q_P}{\Delta T} \tag{50}$$

For 1 mole of an ideal gas:

$$H = U + PV = U + RT \tag{51}$$

Differentiating both sides with respect to temperature T:

$$\frac{dH}{dT} = \frac{dU}{dT} + R\frac{d(T)}{dT} \implies C_p = C_v + R \implies C_p - C_v = R \tag{52}$$

For n moles: $C_{p,\text{total}} - C_{v,\text{total}} = nR$. **Heat Capacity Ratio (γ):**

$$\gamma = \frac{C_p}{C_v} = 1 + \frac{R}{C_v} \tag{53}$$

- **Monoatomic Gas ($\text{He}, \text{Ne}, \text{Ar}$):** Degrees of freedom $f = 3 \implies C_v = \frac{3}{2}R, C_p = \frac{5}{2}R \implies \gamma = \frac{5}{3} \approx 1.67$.
- **Diatomic Gas ($\text{O}_2, \text{N}_2, \text{CO}$ at moderate T):** $f = 5 \implies C_v = \frac{5}{2}R, C_p = \frac{7}{2}R \implies \gamma = \frac{7}{5} = 1.40$.
- **Non-linear Polyatomic Gas ($\text{H}_2\text{O}, \text{NH}_3, \text{CH}_4$):** $f = 6 \implies C_v = 3R, C_p = 4R \implies \gamma = \frac{4}{3} \approx 1.33$.

4.4 Reversible Adiabatic Expansion of an Ideal Gas

In an adiabatic process, no heat enters or leaves the system ($q = 0$). By the First Law:

$$dU = \delta w \implies nC_v dT = -PdV = -\frac{nRT}{V}dV \tag{54}$$

Dividing by nT and separating variables:

$$C_v\frac{dT}{T} = -R\frac{dV}{V} = -(C_p - C_v)\frac{dV}{V} \implies \frac{dT}{T} = -(\gamma - 1)\frac{dV}{V} \tag{55}$$

Integrating from state 1 to state 2:

$$\ln\left(\frac{T_2}{T_1}\right) = -(\gamma-1)\ln\left(\frac{V_2}{V_1}\right) = \ln\left[\left(\frac{V_1}{V_2}\right)^{\gamma-1}\right] \implies T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} = \text{constant} \tag{56}$$

Substituting $T = \frac{PV}{nR}$:

$$P_1V_1^\gamma = P_2V_2^\gamma = \text{constant}, \qquad T_1^\gamma P_1^{1-\gamma} = T_2^\gamma P_2^{1-\gamma} = \text{constant} \tag{57}$$

Work Done in Reversible Adiabatic Expansion:

$$w_{\text{adi}} = \Delta U = nC_v(T_2 - T_1) = \frac{nR(T_2 - T_1)}{\gamma - 1} = \frac{P_2V_2 - P_1V_1}{\gamma - 1} \tag{58}$$

Since the gas expands ($V_2 > V_1$), $T_2 < T_1$ (gas cools during adiabatic expansion!). On a $P-V$ diagram, the slope of an adiabat is γ times steeper than an isotherm:

$$\left(\frac{dP}{dV}\right)_{\text{adi}} = -\gamma\frac{P}{V} = \gamma\left(\frac{dP}{dV}\right)_{\text{iso}} \tag{59}$$

4.5 Thermochemistry: Heats of Reactions, Hess's Law & Cycles

Thermochemical Laws & Enthalpy Formulations

- **Hess's Law of Constant Heat Summation:** If a chemical reaction can be made to take place in a single step or in several successive steps, the overall enthalpy change is identical, independent of the pathway:

$$\Delta_r H^\circ = \Delta H_1 + \Delta H_2 + \Delta H_3 + \dots \tag{60}$$

- **Standard Enthalpy of Formation ($\Delta_f H^\circ$):** Enthalpy change when 1 mole of a compound is synthesized from its constituent elements in their standard reference states (298.15 K, 1 bar).

$$\Delta_r H^\circ = \sum \nu_p \Delta_f H^\circ(\text{products}) - \sum \nu_r \Delta_f H^\circ(\text{reactants}) \tag{61}$$

- **Standard State Convention Trap:** By definition, $\Delta_f H^\circ = 0$ for all pure elements in their most stable allotropic form at 298 K: C(graphite) = 0 (but C(diamond) = $+1.89\,\text{kJ mol}^{-1}$); S(rhombic) = 0 (monoclinic $\neq 0$); P(white) = 0 (red $\neq 0$); $\text{Br}_{2(\text{l})} = 0$; $\text{I}_{2(\text{s})} = 0$.
- **Bond Dissociation Enthalpy (BE):**

$$\Delta_r H^\circ = \sum \text{BE(reactants)} - \sum \text{BE(products)} \quad \text{(all species in gas phase!)} \tag{62}$$

- **Kirchhoff's Law (Temperature Dependence of $\Delta_r H$):**

$$\Delta_r H^\circ_{T_2} = \Delta_r H^\circ_{T_1} + \int_{T_1}^{T_2} \Delta C_p dT = \Delta_r H^\circ_{T_1} + \Delta C_p (T_2 - T_1) \tag{63}$$

where $\Delta C_p = \sum \nu_p C_{p,\text{products}} - \sum \nu_r C_{p,\text{reactants}}$.

Born-Haber Cycle for Ionic Crystal Lattice Enthalpy ($\Delta_{\text{lattice}} H^\circ$)

For crystalline sodium chloride ($\text{NaCl}_{(\text{s})}$), the standard enthalpy of formation from elemental sodium and chlorine gas connects directly to lattice enthalpy via five thermochemical steps:

$$\text{Na}_{(\text{s})} + \frac{1}{2}\text{Cl}_{2(\text{g})} \xrightarrow{\Delta_f H^\circ} \text{NaCl}_{(\text{s})} \tag{64}$$

1. Sublimation of solid sodium: $\text{Na}_{(\text{s})} \to \text{Na}_{(\text{g})}$, $\Delta H^\circ_{\text{sub}} = +108.4\,\text{kJ mol}^{-1}$.
2. Ionization of gaseous sodium: $\text{Na}_{(\text{g})} \to \text{Na}^+_{(\text{g})} + e^-$, $\text{IE}_1 = +495.8\,\text{kJ mol}^{-1}$.
3. Dissociation of diatomic chlorine: $\frac{1}{2}\text{Cl}_{2(\text{g})} \to \text{Cl}_{(\text{g})}$, $\frac{1}{2}\Delta_{\text{diss}} H^\circ = +\frac{1}{2}(242) = +121.0\,\text{kJ mol}^{-1}$.
4. Electron gain by chlorine: $\text{Cl}_{(\text{g})} + e^- \to \text{Cl}^-_{(\text{g})}$, $\Delta_{\text{eg}} H^\circ = -348.6\,\text{kJ mol}^{-1}$.
5. Condensation of gaseous ions into crystal lattice: $\text{Na}^+_{(\text{g})} + \text{Cl}^-_{(\text{g})} \to \text{NaCl}_{(\text{s})}$, $-\Delta_{\text{lattice}} H^\circ$.

Applying Hess's Law:

$$\Delta_f H^\circ = \Delta H^\circ_{\text{sub}} + \text{IE}_1 + \frac{1}{2}\Delta_{\text{diss}} H^\circ + \Delta_{\text{eg}} H^\circ - \Delta_{\text{lattice}} H^\circ \tag{65}$$

$$\Delta_{\text{lattice}} H^\circ = \Delta H^\circ_{\text{sub}} + \text{IE}_1 + \frac{1}{2}\Delta_{\text{diss}} H^\circ + \Delta_{\text{eg}} H^\circ - \Delta_f H^\circ = +788\,\text{kJ mol}^{-1} \tag{66}$$

4.6 Second & Third Laws: Entropy, Free Energy & Spontaneity

- **Entropy (S):** Measure of microscopic disorder or thermal energy dispersal. Infinitesimal reversible change:

$$dS = \frac{\delta q_{\text{rev}}}{T} \tag{67}$$

- **Entropy Change for an Ideal Gas:**

$$\Delta S = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right) = nC_p \ln\left(\frac{T_2}{T_1}\right) - nR \ln\left(\frac{P_2}{P_1}\right) \tag{68}$$

  - **Isothermal Process ($T_1 = T_2$):** $\Delta S = nR\ln(V_2/V_1) = nR\ln(P_1/P_2)$.
  - **Isobaric Process ($P_1 = P_2$):** $\Delta S = nC_p\ln(T_2/T_1)$.
  - **Isochoric Process ($V_1 = V_2$):** $\Delta S = nC_v\ln(T_2/T_1)$.
  - **Reversible Phase Transitions:** $\Delta S_{\text{fusion}} = \frac{\Delta H_{\text{fus}}}{T_f}$, $\Delta S_{\text{vaporization}} = \frac{\Delta H_{\text{vap}}}{T_b}$.
- **Spontaneity Criterion (Second Law of Thermodynamics):** For an isolated system or the universe:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} \geq 0 \quad (\text{Spontaneous if } > 0; \text{Equilibrium if } = 0) \tag{69}$$

- **Third Law of Thermodynamics:** The entropy of a perfectly crystalline pure substance approaches zero as absolute temperature approaches zero: $\lim_{T\to 0} S = 0$.

4.7 Gibbs Free Energy & Equilibrium Thermodynamics

Gibbs Free Energy ($G = H - TS$) provides the single-parameter spontaneity criterion at constant temperature and pressure:

$$\Delta G = \Delta H - T\Delta S \tag{70}$$

- $\Delta G < 0 \implies$ Reaction is spontaneous in the forward direction.
- $\Delta G = 0 \implies$ Dynamic equilibrium is attained.
- $\Delta G > 0 \implies$ Non-spontaneous forward (spontaneous in reverse direction).

ΔH	ΔS	$\Delta G = \Delta H - T\Delta S$	**Spontaneity Regime & Temperature Dependence**
$-$	$+$	Always Negative ($-$)	**Spontaneous at all temperatures** (enthalpy & entropy driven)
$+$	$-$	Always Positive ($+$)	**Non-spontaneous at all temperatures** (reverse is spontaneous)
$-$	$-$	Negative at low T, Positive at high T	**Spontaneous at low** T, non-spontaneous at high T ($\lvert T\Delta S\rvert > \lvert\Delta H\rvert$)
$+$	$+$	Positive at low T, Negative at high T	**Spontaneous at high** T, non-spontaneous at low T ($T > \frac{\Delta H}{\Delta S}$)

Table 3: Thermodynamic Spontaneity Temperature Dependency Matrix.

Master Thermodynamic Equilibrium Links

For a general reaction with reaction quotient Q:

$$\Delta G = \Delta_r G^\circ + RT\ln Q = \Delta_r G^\circ + 2.303RT\log_{10} Q \tag{71}$$

At dynamic chemical equilibrium, $\Delta G = 0$ and $Q = K_{\text{eq}}$:

$$\Delta_r G^\circ = -RT\ln K_{\text{eq}} = -2.303RT\log_{10} K_{\text{eq}} \iff K_{\text{eq}} = e^{-\Delta_r G^\circ/RT} \tag{72}$$

van 't Hoff Reaction Isobar:

$$\frac{d\ln K}{dT} = \frac{\Delta_r H^\circ}{RT^2} \implies \log_{10}\left(\frac{K_2}{K_1}\right) = \frac{\Delta_r H^\circ}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] = \frac{\Delta_r H^\circ}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \tag{73}$$

- **Endothermic Reaction ($\Delta_r H^\circ > 0$):** K increases as temperature increases.
- **Exothermic Reaction ($\Delta_r H^\circ < 0$):** K decreases as temperature increases.

5 Chemical & Ionic Equilibrium (Unit 6)

5.1 Chemical Equilibrium: Law of Mass Action & Constant Relations

For a reversible homogeneous gas-phase reaction:

$$aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g) \tag{74}$$

The equilibrium constants in terms of molar concentrations (K_c) and partial pressures (K_p) are:

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}, \qquad K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b} \tag{75}$$

Derivation of $K_p = K_c(RT)^{\Delta n_g}$

Assuming ideal gas behavior for each component:

$$P_i V = n_i RT \implies P_i = \left(\frac{n_i}{V}\right) RT = [i]RT \tag{76}$$

Substituting each partial pressure into the K_p expression:

$$K_p = \frac{([C]RT)^c([D]RT)^d}{([A]RT)^a([B]RT)^b} = \frac{[C]^c[D]^d}{[A]^a[B]^b} \times \frac{(RT)^{c+d}}{(RT)^{a+b}} = K_c(RT)^{(c+d)-(a+b)} \tag{77}$$

$$K_p = K_c(RT)^{\Delta n_g} \tag{78}$$

where $\Delta n_g = (c+d) - (a+b)$.

- If $\Delta n_g = 0 \implies K_p = K_c$ (dimensionless, pressure independent, e.g., $H_2 + I_2 \rightleftharpoons 2HI$).
- If $\Delta n_g > 0 \implies K_p > K_c$ when $RT > 1$ (units $\text{atm}^{\Delta n_g}$ or $\text{bar}^{\Delta n_g}$, e.g., $PCl_5 \rightleftharpoons PCl_3 + Cl_2$).
- If $\Delta n_g < 0 \implies K_p < K_c$ when $RT > 1$ (e.g., $N_2 + 3H_2 \rightleftharpoons 2NH_3$).

5.2 Le Chatelier's Principle & Equilibrium Shifts

If a dynamic equilibrium is subjected to a stress in concentration, pressure, volume, temperature, or inert gas, the system shifts in that direction which minimizes or undoes the applied constraint:

- **Concentration:** Adding reactants or removing products shifts forward ($Q < K$).
- **Pressure / Volume:** Increasing pressure (compressing volume) shifts equilibrium toward fewer gaseous moles ($\Delta n_g < 0$).
- **Temperature:** Increasing temperature shifts toward the endothermic direction ($\Delta H > 0$).
- **Inert Gas Addition:**
  - **At Constant Volume:** Total pressure rises, but partial pressures and molar concentrations of reacting gases remain **strictly unchanged** $\implies$ **NO EFFECT** on equilibrium state!
  - **At Constant Pressure:** Volume expands, partial pressures drop $\implies$ equilibrium shifts toward **greater number of gaseous moles** ($\Delta n_g > 0$).
- **Catalyst:** Increases forward and backward rates equally; equilibrium constant K and composition are completely unaffected (shortens time to achieve equilibrium).

NCERT Focus: Le Chatelier Optimization in Industrial Syntheses

Case Study: Synthesis of Ammonia (Haber-Bosch Process)

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \quad \Delta_r H^\circ = -92.4\,\text{kJ}\cdot\text{mol}^{-1}, \quad \Delta n_g = 2 - 4 = -2$$

- **High Pressure ($\approx 200\,\text{atm}$):** Since $\Delta n_g < 0$, high pressure drives the equilibrium forward toward product formation, optimizing ammonia yield.
- **Optimum Temperature (≈ 700–$775\,\text{K}$):** Because the reaction is exothermic ($\Delta_r H^\circ < 0$), low temperature thermodynamically favors ammonia, but drastically lowers kinetic reaction rates. An optimum temperature with iron oxide catalyst and promoters (K_2O and Al_2O_3) is maintained.
- **Inert Gas Injection Trap:** Adding argon or neon at constant volume ($V = \text{const}$) causes zero shift in equilibrium yield. Only addition at constant pressure ($P = \text{const}$) shifts the reaction backward (toward more gaseous moles, $\Delta n_g = -2 < 0 \implies$ leftward shift).

5.3 Degree of Dissociation from Vapour Density

For a gas dissociating according to $A(g) \rightleftharpoons nB(g)$:

$$\alpha = \frac{D-d}{(n-1)d} = \frac{M_{\text{theoretical}} - M_{\text{observed}}}{(n-1)M_{\text{observed}}} \tag{79}$$

where D is the theoretical vapour density of pure reactant, and d is the observed equilibrium mixture vapour density.

5.4 Ionic Equilibrium: Ostwald Dilution Law & Autoionization of Water

- **Ostwald's Dilution Law for Weak Acid** ($\mathrm{HA} \rightleftharpoons \mathrm{H^+ + A^-}$)**:**

$$K_a = \frac{C\alpha^2}{1-\alpha} \approx C\alpha^2 \quad (\text{for } \alpha \leq 0.05) \implies \alpha = \sqrt{\frac{K_a}{C}} \tag{80}$$

$$[\mathrm{H^+}] = C\alpha = \sqrt{K_a C} \implies \mathrm{pH} = \frac{1}{2}\left(\mathrm{p}K_a - \log_{10} C\right) \tag{81}$$

- **Self-Ionization of Water** ($2\mathrm{H_2O} \rightleftharpoons \mathrm{H_3O^+ + OH^-}$)**:**

$$K_w = [\mathrm{H^+}][\mathrm{OH^-}] = 1.0 \times 10^{-14} \quad (\text{at } 25^\circ\mathrm{C}) \implies \mathrm{pH} + \mathrm{pOH} = \mathrm{p}K_w = 14 \tag{82}$$

  Because self-ionization is endothermic ($\Delta H > 0$), K_w increases with temperature: at 80°C, $K_w \approx 10^{-13} \implies$ pH of neutral water $= \frac{13}{2} = 6.5$.

5.5 Salt Hydrolysis: Derivations of K_h, h, and pH

Salt hydrolysis is the reverse of neutralization, where ions of a salt interact with water to perturb $[\mathrm{H^+}]$ and $[\mathrm{OH^-}]$:

Salt Category & Example	**Hydrolysis Type**	**Hydrolysis Constant (K_h)**	**Degree of Hydrolysis (h)**	**Exact pH Formula**
Strong Acid + Strong Base ($\mathrm{NaCl}, \mathrm{KNO_3}, \mathrm{NaClO_4}$)	No Hydrolysis Spectator ions	—	$h = 0$	$\mathrm{pH} = 7.00$ (Neutral, unaffected by dilution)
Strong Acid + Weak Base ($\mathrm{NH_4Cl}, \mathrm{FeSO_4}, \mathrm{AlCl_3}$)	Cationic	$K_h = \frac{K_w}{K_b}$	$h = \sqrt{\frac{K_w}{K_b C}}$	$\mathrm{pH} = 7 - \frac{1}{2}\mathrm{p}K_b - \frac{1}{2}\log_{10} C$ (Acidic: pH < 7)
Weak Acid + Strong Base ($\mathrm{CH_3COONa}, \mathrm{NaCN}, \mathrm{Na_2CO_3}$)	Anionic	$K_h = \frac{K_w}{K_a}$	$h = \sqrt{\frac{K_w}{K_a C}}$	$\mathrm{pH} = 7 + \frac{1}{2}\mathrm{p}K_a + \frac{1}{2}\log_{10} C$ (Alkaline: pH > 7)
Weak Acid + Weak Base ($\mathrm{CH_3COONH_4}, \mathrm{NH_4CN}$)	Both Ions Hydrolyze	$K_h = \frac{K_w}{K_a K_b}$	$h = \sqrt{\frac{K_w}{K_a K_b}}$ **(Independent of C!)**	$\mathrm{pH} = \mathbf{7 + \frac{1}{2}pK_a - \frac{1}{2}pK_b}$ **(Strictly independent of C!)**

Table 4: Master Classification and Analytical Formulations for Salt Hydrolysis.

5.6 Buffer Solutions & Henderson-Hasselbalch Derivations

A buffer resists pH changes upon addition of small quantities of strong acid or base.

1. **Acidic Buffer (Weak Acid + Salt with Strong Base, e.g.,** $\mathrm{CH_3COOH + CH_3COONa}$**):**

$$K_a = \frac{[\mathrm{H^+}][\mathrm{A^-}]}{[\mathrm{HA}]} \implies [\mathrm{H^+}] = K_a \frac{[\mathrm{HA}]}{[\mathrm{A^-}]} \approx K_a \frac{[\text{Acid}]}{[\text{Salt}]} \tag{83}$$

Taking $-\log_{10}$ on both sides:

$$\text{pH} = \text{p}K_a + \log_{10}\left(\frac{[\text{Conjugate Base / Salt}]}{[\text{Weak Acid}]}\right) \tag{84}$$

2. **Basic Buffer (Weak Base + Salt with Strong Acid, e.g., $NH_4OH + NH_4Cl$):**

$$\text{pOH} = \text{p}K_b + \log_{10}\left(\frac{[\text{Conjugate Acid / Salt}]}{[\text{Weak Base}]}\right), \qquad \text{pH} = 14 - \text{pOH} \tag{85}$$

3. **Buffer Capacity (ϕ):** The number of moles of strong acid or base required to change the pH of 1 L of buffer solution by 1 unit:

$$\phi = \frac{dB}{d(\text{pH})} = 2.303\frac{[\text{Acid}][\text{Salt}]}{[\text{Acid}] + [\text{Salt}]} \tag{86}$$

Buffer capacity is **maximum** when $[\text{Acid}] = [\text{Salt}] \implies \text{pH} = \text{p}K_a$ (effective buffer range: $\text{p}K_a \pm 1$).

5.7 Solubility Product (K_{sp}) & Common Ion Precipitation

For a sparingly soluble salt A_xB_y:

$$A_xB_y(s) \rightleftharpoons xA^{y+}(aq) + yB^{x-}(aq) \tag{87}$$

If molar solubility is $S\,\text{mol}\,\text{L}^{-1}$:

$$[A^{y+}] = xS, \qquad [B^{x-}] = yS \tag{88}$$

$$K_{sp} = [A^{y+}]^x[B^{x-}]^y = (xS)^x(yS)^y = x^xy^yS^{x+y} \implies S = \left(\frac{K_{sp}}{x^xy^y}\right)^{\frac{1}{x+y}} \tag{89}$$

Solubility Equations by Salt Stoichiometry:

- AB type ($AgCl, BaSO_4$): $K_{sp} = S^2 \implies S = \sqrt{K_{sp}}$.
- AB_2 or A_2B type ($PbCl_2, Ag_2CrO_4, CaF_2$): $K_{sp} = 1^1 \times 2^2 \times S^3 = 4S^3 \implies S = \sqrt[3]{K_{sp}/4}$.
- AB_3 or A_3B type ($Al(OH)_3, Fe(OH)_3$): $K_{sp} = 1^1 \times 3^3 \times S^4 = 27S^4 \implies S = \sqrt[4]{K_{sp}/27}$.
- A_2B_3 or A_3B_2 type ($As_2S_3, Ca_3(PO_4)_2$): $K_{sp} = 2^2 \times 3^3 \times S^5 = 108S^5 \implies S = \sqrt[5]{K_{sp}/108}$.

Precipitation Criterion:

- $Q_{sp} < K_{sp} \implies$ Unsaturated solution; no precipitate forms.
- $Q_{sp} = K_{sp} \implies$ Saturated solution at dynamic equilibrium.
- $Q_{sp} > K_{sp} \implies$ Supersaturated solution; **precipitation occurs immediately**.

6 Solutions & Colligative Properties (Unit 5)

6.1 Gas Solubility & Henry's Law

Henry's Law states that the solubility of a gas in a liquid at constant temperature is directly proportional to the partial pressure of the gas above the solution:

$$p = K_H \cdot x \iff x = \frac{p}{K_H} \tag{90}$$

where p is the partial pressure of gas, x is the mole fraction of gas dissolved, and K_H is Henry's law constant.

- At a constant pressure p, solubility x is **inversely proportional** to K_H.
- For a given gas, K_H **increases with increasing temperature** $\implies$ gas solubility in liquids decreases on heating (aquatic life thrives in cold water due to higher dissolved O_2).

6.2 Vapour Pressure & Raoult's Law

For a binary solution of volatile liquids A and B, Raoult's law states that the partial vapour pressure of each component is equal to the vapour pressure of the pure component multiplied by its mole fraction in the liquid phase:

$$P_A = P_A^\circ x_A, \qquad P_B = P_B^\circ x_B \tag{91}$$

By Dalton's Law of Partial Pressures, the total vapour pressure is:

$$P_{\text{total}} = P_A + P_B = P_A^\circ x_A + P_B^\circ x_B = P_A^\circ(1 - x_B) + P_B^\circ x_B = P_A^\circ + (P_B^\circ - P_A^\circ)x_B \tag{92}$$

Vapour Phase Composition (y_A, y_B):

$$y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^\circ x_A}{P_{\text{total}}}, \qquad y_B = \frac{P_B}{P_{\text{total}}} = \frac{P_B^\circ x_B}{P_{\text{total}}} \implies \frac{1}{P_{\text{total}}} = \frac{y_A}{P_A^\circ} + \frac{y_B}{P_B^\circ} \tag{93}$$

6.3 Ideal vs Non-Ideal Solutions & Azeotropes

Thermodynamic Property	Ideal Solution	Non-Ideal: Positive Deviation	Non-Ideal: Negative Deviation
Intermolecular Forces	$A-B \approx A-A \approx B-B$	$A-B < A-A, B-B$ (Weaker)	$A-B > A-A, B-B$ (Stronger)
Vapour Pressure	$P_{\text{total}} = P_A^\circ x_A + P_B^\circ x_B$	$P_{\text{total}} > P_{\text{ideal}}$	$P_{\text{total}} < P_{\text{ideal}}$
Enthalpy of Mixing	$\Delta H_{\text{mix}} = 0$	$\Delta H_{\text{mix}} > 0$ (Endothermic)	$\Delta H_{\text{mix}} < 0$ (Exothermic)
Volume of Mixing	$\Delta V_{\text{mix}} = 0$	$\Delta V_{\text{mix}} > 0$ (Expansion)	$\Delta V_{\text{mix}} < 0$ (Contraction)
Entropy & Free Energy	$\Delta S_{\text{mix}} > 0, \Delta G_{\text{mix}} < 0$	$\Delta S_{\text{mix}} > 0, \Delta G_{\text{mix}} < 0$	$\Delta S_{\text{mix}} > 0, \Delta G_{\text{mix}} < 0$
Azeotrope Formed	None (Separable by distillation)	**Minimum Boiling Azeotrope**	**Maximum Boiling Azeotrope**
High-Yield Examples	Benzene + Toluene; n-hexane + n-heptane; $C_2H_5Br + C_2H_5I$	Ethanol + Water (95.6%); Acetone + CS_2; CCl_4 + Toluene	Chloroform + Acetone; $HNO_3 + H_2O$ (68%); $HCl + H_2O$ (20.2%)

Table 5: Comprehensive Physical and Thermodynamic Profiling of Binary Liquid Mixtures.

6.4 Colligative Properties & Thermodynamic Derivations

Colligative properties depend exclusively on the ratio of the number of solute particles to solvent particles, irrespective of their chemical nature.

6.4.1 1. Relative Lowering of Vapour Pressure (RLVP)

When a non-volatile solute B is added to volatile solvent A, only solvent molecules occupy the vapour phase: $P_{\text{total}} = P_A = P_A^\circ x_A = P_A^\circ(1 - x_B)$.

$$\Delta P = P_A^\circ - P = P_A^\circ x_B \implies \frac{P_A^\circ - P}{P_A^\circ} = x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{w_B M_A}{M_B w_A} \quad \text{(dilute)} \tag{94}$$

Exact Molality Expression for All Concentrations:

$$\frac{P_A^\circ - P}{P} = \frac{n_B}{n_A} = \frac{m \times M_A}{1000} \implies \text{Notice denominator is } P, \text{ not } P_A^\circ! \tag{95}$$

Common Pitfall: Relative Lowering of Vapour Pressure Denominator

A frequent numerical trap in NEET involves confusing the theoretical RLVP definition with the exact molality formula:

- **Definition of RLVP:** By Raoult's law, relative lowering with respect to pure solvent vapour pressure is:

$$\frac{P^\circ - P}{P^\circ} = x_B = \frac{n_B}{n_A + n_B}$$

  This equation strictly possesses the pure solvent vapour pressure P° in the denominator.
- **Exact Non-Approximated Molality Conversion:** Inverting and subtracting 1 gives:

$$\frac{P^\circ - P}{P} = \frac{n_B}{n_A} = \frac{w_B \cdot M_A}{M_B \cdot w_A} = \frac{m \cdot M_A}{1000}$$

  Here the denominator is the **solution** vapour pressure P, NOT P°. Equating $(P^\circ - P)/P^\circ$ directly to n_B/n_A without dilute approximation creates systematic calculation errors in non-dilute questions ($m > 0.1\,\text{m}$).

6.4.2 2. Elevation of Boiling Point (ΔT_b)

At boiling point, vapour pressure equals external atmospheric pressure (1 atm). Addition of non-volatile solute lowers vapour pressure; hence the solution must be heated to a higher temperature $T_b > T_b^\circ$:

$$\Delta T_b = T_b - T_b^\circ = K_b \cdot m = \frac{1000 K_b w_B}{M_B w_A} \tag{96}$$

Thermodynamic Derivation of Ebullioscopic Constant (K_b): From the Clausius-Clapeyron relation connecting chemical potential and vapour pressure:

$$\ln\left(\frac{P^\circ}{P}\right) = \frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_b^\circ} - \frac{1}{T_b}\right) \approx \frac{\Delta H_{\text{vap}} \Delta T_b}{R(T_b^\circ)^2} \tag{97}$$

Since $\ln(P^\circ/P) = -\ln(1 - x_B) \approx x_B \approx \frac{mM_A}{1000}$:

$$\frac{mM_A}{1000} = \frac{\Delta H_{\text{vap}} \Delta T_b}{R(T_b^\circ)^2} \implies \Delta T_b = \left[\frac{R(T_b^\circ)^2 M_A}{1000 \Delta H_{\text{vap}}}\right] m \tag{98}$$

$$K_b = \frac{R(T_b^\circ)^2 M_A}{1000 \Delta H_{\text{vap}}} = \frac{R(T_b^\circ)^2}{1000 l_v} \quad (l_v = \text{latent heat of vaporization per gram}) \tag{99}$$

For water: $K_b = 0.512 \approx 0.52\,\text{K kg mol}^{-1}$.

6.4.3 3. Depression of Freezing Point (ΔT_f)

Freezing point is the temperature at which the vapour pressure of the liquid solvent equals the vapour pressure of its solid phase. Lowering vapour pressure depresses the freezing point:

$$\Delta T_f = T_f^\circ - T_f = K_f \cdot m = \frac{1000 K_f w_B}{M_B w_A} \tag{100}$$

Thermodynamic Derivation of Cryoscopic Constant (K_f):

$$K_f = \frac{R(T_f^\circ)^2 M_A}{1000 \Delta H_{\text{fusion}}} = \frac{R(T_f^\circ)^2}{1000 l_f} \quad (l_f = \text{latent heat of fusion per gram}) \tag{101}$$

For water: $K_f = 1.86\,\text{K kg mol}^{-1}$. Note that $K_f/K_b = 1.86/0.52 \approx 3.58$ (freezing point depression is over 3.5 times more sensitive than boiling point elevation!).

NEET High-Yield Application: Automotive Antifreeze Formulation

Problem Statement: An automobile radiator requires protection against freezing down to $-15.0\,^\circ\text{C}$ during harsh winter operations. Calculate the mass of ethylene glycol ($C_2H_6O_2$, molar mass $M = 62.07\,\text{g}\cdot\text{mol}^{-1}$) that must be added to 5.00 kg of water ($K_f = 1.86\,\text{K}\cdot\text{kg}\cdot\text{mol}^{-1}$).

1. **Target Freezing Point Depression:**

$$\Delta T_f = T_f^\circ - T_f = 0.00\,^\circ\text{C} - (-15.0\,^\circ\text{C}) = 15.0\,\text{K}$$

2. **Required Molality of Glycol:**

$$m = \frac{\Delta T_f}{K_f} = \frac{15.0\,\text{K}}{1.86\,\text{K}\cdot\text{kg}\cdot\text{mol}^{-1}} = 8.065\,\text{mol}\cdot\text{kg}^{-1}$$

3. **Mass of Ethylene Glycol Required:**

$$w_{\text{glycol}} = m \times W_{\text{water}}\ (\text{in kg}) \times M_{\text{glycol}} = 8.065\,\text{mol}\cdot\text{kg}^{-1} \times 5.00\,\text{kg} \times 62.07\,\text{g}\cdot\text{mol}^{-1} \approx 2503\,\text{g} = 2.50\,\text{kg}$$

Clinical / Engineering Context: In addition to lowering the freezing point, the non-volatile ethylene glycol simultaneously elevates the boiling point ($\Delta T_b = K_b \cdot m = 0.52 \times 8.065 \approx 4.19\,\text{K}$), preventing engine coolant boilover during high-temperature engine cycling.

6.4.4 4. Osmotic Pressure (Π)

Osmosis is the spontaneous net flow of solvent molecules through a semi-permeable membrane from pure solvent into solution. Osmotic pressure Π is the excess hydrostatic pressure that must be applied to the solution to completely arrest this solvent influx:

$$\Pi = CRT = \left(\frac{n_B}{V}\right)RT = \frac{w_B RT}{M_B V} \implies M_B = \frac{w_B RT}{\Pi V} \tag{102}$$

Why Osmotic Pressure is Superior for Macromolecules (Proteins, DNA, Polymers):

1. High molecular mass ($M_B \sim 10^4 - 10^6\,\text{g mol}^{-1}$) produces immeasurably tiny molalities ($m \sim 10^{-4}\,\text{m}$), causing negligible ΔT_b ($< 10^{-4}\,\text{K}$), but yielding easily measurable osmotic pressure ($\Pi = 10^{-4} \times 0.0821 \times 300 \approx 2.5 \times 10^{-3}\,\text{atm} \approx 26\,\text{mm Hg}$).
2. Measurement occurs at ambient room temperature, preventing thermal denaturation of delicate biological polymers.

6.5 Abnormal Molar Masses & van 't Hoff Factor (i)

When a solute undergoes dissociation or association in solution, colligative properties deviate from theoretical ideal values:

$$i = \frac{\text{Experimental / Observed Colligative Property}}{\text{Calculated / Normal Colligative Property}} = \frac{M_{\text{theoretical / normal}}}{M_{\text{observed / abnormal}}} \tag{103}$$

Derivation of Degree of Dissociation (α_{diss}) and Association (α_{assoc})

- **Case 1: Dissociation ($A \rightleftharpoons nB$):** Let 1 mole of electrolyte be dissolved with degree of dissociation α:

$$\begin{array}{lccc} & A & \rightleftharpoons & nB \\ \text{Initial moles:} & 1 & & 0 \\ \text{Equilibrium moles:} & 1-\alpha & & n\alpha \end{array} \implies \text{Total moles} = 1 - \alpha + n\alpha = 1 + (n-1)\alpha \tag{104}$$

$$i = \frac{1 + (n-1)\alpha}{1} \implies \alpha_{\text{diss}} = \frac{i-1}{n-1} \tag{105}$$

- **Case 2: Association / Polymerization ($nA \rightleftharpoons A_n$):** Let 1 mole of solute undergo association to an n-mer with degree of association α:

$$\begin{array}{lccc} & nA & \rightleftharpoons & A_n \\ \text{Initial moles:} & 1 & & 0 \\ \text{Equilibrium moles:} & 1-\alpha & & \frac{\alpha}{n} \end{array} \implies \text{Total moles} = 1 - \alpha + \tfrac{\alpha}{n} = 1 - \left(1 - \tfrac{1}{n}\right)\alpha \tag{106}$$

$$i = 1 - \left(1 - \frac{1}{n}\right)\alpha \implies \alpha_{\text{assoc}} = \frac{1-i}{1-\frac{1}{n}} \tag{107}$$

For dimerization of acetic acid in benzene ($n = 2$): $i = 1 - \frac{\alpha}{2} \implies$ if 100% dimerized ($\alpha = 1$), $i = 0.5$ and observed molar mass is double ($120\,\text{g mol}^{-1}$ vs $60\,\text{g mol}^{-1}$).

7 Redox Reactions & Electrochemistry (Unit 7)

7.1 Galvanic Cells & The Daniell Cell

A galvanic cell converts spontaneous chemical redox free energy into electrical work:

$$\text{Zn(s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)} \tag{108}$$

- **Anode (Oxidation, Negative Terminal):** $\text{Zn(s)} \rightarrow \text{Zn}^{2+}\text{(aq)} + 2e^-$, $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76\,\text{V}$.
- **Cathode (Reduction, Positive Terminal):** $\text{Cu}^{2+}\text{(aq)} + 2e^- \rightarrow \text{Cu(s)}$, $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34\,\text{V}$.
- **Cell Representation:** $\text{Zn(s)} \mid \text{Zn}^{2+}\text{(aq)} \parallel \text{Cu}^{2+}\text{(aq)} \mid \text{Cu(s)}$.
- **Salt Bridge Functions:**
  1. Maintains electrical neutrality in both half-cells by migrating counter-ions (K^+ and Cl^- or NO_3^-).
  2. Prevents physical mixing of the two electrolytic solutions.
  3. Minimizes liquid junction potential. Electrolyte chosen ($\text{KCl}, \text{KNO}_3, \text{NH}_4\text{NO}_3$) has nearly identical transport numbers for cation and anion ($v_+ \approx v_-$).

7.2 Standard Hydrogen Electrode & Electrochemical Series

The Standard Hydrogen Electrode (SHE) is the universal reference standard assigned $E^\circ = 0.000\,\text{V}$ at all temperatures:

$$\text{Pt(s)} \mid \text{H}_2\text{(g, 1 bar)} \mid \text{H}^+\text{(aq, 1 M)}, \qquad 2\text{H}^+ + 2e^- \rightleftharpoons \text{H}_2\text{(g)} \tag{109}$$

Standard Cell EMF (E°_{cell}):

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}} = E^\circ_{\text{right}} - E^\circ_{\text{left}} \quad \text{(using standard reduction potentials)} \tag{110}$$

7.3 Thermodynamic Derivation of the Nernst Equation

The maximum non-expansion electrical work delivered by a galvanic cell at constant T, P equals the decrease in Gibbs free energy:

$$w_{\text{elec}} = q_{\text{charge}} \times E_{\text{cell}} = (nF)E_{\text{cell}} \implies \Delta_r G = -w_{\text{elec}} = -nFE_{\text{cell}} \tag{111}$$

Under standard state conditions: $\Delta_r G^\circ = -nFE^\circ_{\text{cell}}$. Substituting these relations into the fundamental thermodynamic equation $\Delta_r G = \Delta_r G^\circ + RT\ln Q$:

$$-nFE_{\text{cell}} = -nFE^\circ_{\text{cell}} + RT\ln Q \tag{112}$$

Dividing throughout by $-nF$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{RT}{nF}\ln Q = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log_{10} Q \tag{113}$$

Substituting universal constants $R = 8.314\,\text{J mol}^{-1}\,\text{K}^{-1}$, $F = 96485\,\text{C mol}^{-1}$, and $T = 298.15\,\text{K}$:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} Q \tag{114}$$

Nernst Equation Special Cases & Thermodynamic Properties

- **Dynamic Equilibrium ($E_{\text{cell}} = 0, Q = K_c$):**

$$0 = E^\circ_{\text{cell}} - \frac{0.0591}{n}\log_{10} K_c \implies \log_{10} K_c = \frac{nE^\circ_{\text{cell}}}{0.0591} \tag{115}$$

- **Concentration Cells (Identical Electrodes, Different Concentrations $C_1 < C_2$):** Since standard potentials are identical, $E^\circ_{\text{cell}} = 0$:

$$E_{\text{cell}} = -\frac{0.0591}{n}\log_{10}\left(\frac{C_1}{C_2}\right) = \frac{0.0591}{n}\log_{10}\left(\frac{C_2}{C_1}\right) \quad \text{(Spontaneous if } C_2 > C_1\text{)} \tag{116}$$

- **Temperature Coefficient of EMF and Entropy of Cell Reaction:**

$$\left(\frac{\partial \Delta G}{\partial T}\right)_P = -\Delta S \implies \Delta S = nF\left(\frac{\partial E_{\text{cell}}}{\partial T}\right)_P \tag{117}$$

$$\Delta H = \Delta G + T\Delta S = -nFE_{\text{cell}} + nFT\left(\frac{\partial E_{\text{cell}}}{\partial T}\right)_P \tag{118}$$

7.4 Electrolytic Conductance & Kohlrausch's Law

- **Conductivity (κ):** Conductance of a unit volume ($1\,\text{cm}^3$ or $1\,\text{m}^3$) of electrolytic solution:

$$R = \rho\frac{l}{A} \implies G = \frac{1}{R} = \kappa\frac{A}{l} \implies \kappa = \frac{1}{R}\left(\frac{l}{A}\right) = G \cdot G^* \tag{119}$$

  where $G^* = l/A$ is the cell constant (units cm^{-1} or m^{-1}).
- **Molar Conductivity (Λ_m):** Conductance of volume of solution containing 1 mole of electrolyte:

$$\Lambda_m = \frac{\kappa \times 1000}{M \text{ (in Molarity)}} \quad (\text{in S cm}^2\,\text{mol}^{-1}) = \frac{\kappa}{1000 \times M} \quad (\text{in S m}^2\,\text{mol}^{-1}) \tag{120}$$

- **Dilution Effects:**
  - **Conductivity (κ) Decreases with Dilution:** Dilution reduces the number of current-carrying ions per unit volume.
  - **Molar Conductivity (Λ_m) Increases with Dilution:** Volume containing 1 mole expands, overcompensating the decrease in κ.
- **Strong vs Weak Electrolytes:**
  - **Strong Electrolytes:** Obey Debye-Hückel-Onsager equation: $\Lambda_m = \Lambda^\circ_m - A\sqrt{C}$. Plot of Λ_m vs $\sqrt{C}$ is linear and can be extrapolated to $C = 0$ to obtain limiting molar conductivity Λ°_m.
  - **Weak Electrolytes (CH_3COOH):** Λ_m increases steeply near infinite dilution; cannot be obtained by graphical extrapolation.

Kohlrausch's Law of Independent Migration of Ions

At infinite dilution, each ion migrates independently of its co-ion and makes a definite contribution to the total molar conductivity:

$$\Lambda^\circ_m = \nu_+\lambda^\circ_+ + \nu_-\lambda^\circ_- \tag{121}$$

High-Yield Applications of Kohlrausch's Law:

1. **Limiting Molar Conductivity of Weak Electrolytes (CH_3COOH):**

$$\Lambda^\circ_m(CH_3COOH) = \Lambda^\circ_m(CH_3COONa) + \Lambda^\circ_m(HCl) - \Lambda^\circ_m(NaCl) \tag{122}$$

2. **Degree of Dissociation of Weak Electrolyte (α):**

$$\alpha = \frac{\Lambda_m}{\Lambda^\circ_m} \implies K_a = \frac{C\alpha^2}{1-\alpha} = \frac{C\left(\frac{\Lambda_m}{\Lambda^\circ_m}\right)^2}{1-\frac{\Lambda_m}{\Lambda^\circ_m}} = \frac{C\Lambda_m^2}{\Lambda^\circ_m(\Lambda^\circ_m - \Lambda_m)} \tag{123}$$

3. **Solubility and K_{sp} of Sparingly Soluble Salts ($AgCl, BaSO_4$):**

$$\Lambda_m^\circ = \frac{1000\kappa_{\text{salt}}}{S} \implies S = \frac{1000\kappa_{\text{salt}}}{\Lambda_m^\circ} \implies K_{sp} = S^2 \tag{124}$$

where $\kappa_{\text{salt}} = \kappa_{\text{solution}} - \kappa_{\text{water}}$.

7.5 Electrolysis & Faraday's Laws

- **Faraday's First Law:** Mass of substance deposited or liberated at any electrode is directly proportional to the total electrical charge passed:

$$w = Z \cdot Q = Z \cdot I \cdot t = \left(\frac{E}{96500}\right) I \cdot t = \left(\frac{M}{nF}\right) I \cdot t \tag{125}$$

where $Z = \frac{E}{F}$ is the electrochemical equivalent, and $E = \frac{M}{n}$ is equivalent mass.
- **Faraday's Second Law:** When the identical quantity of electricity is passed through different electrolytes connected in series, masses deposited are proportional to their chemical equivalent masses:

$$\frac{w_1}{w_2} = \frac{E_1}{E_2} \tag{126}$$

- **Preferential Discharge Traps:**
  - **Molten** NaCl**:** Cathode: $Na^+ + e^- \to Na(l)$; Anode: $2Cl^- \to Cl_2(g) + 2e^-$.
  - **Aqueous** NaCl**:** At cathode, H_2O reduction ($E^\circ = -0.83\,V$) occurs preferentially over Na^+ ($E^\circ = -2.71\,V$) $\implies$ $H_2(g)$ evolved. At anode, oxidation of Cl^- to $Cl_2(g)$ occurs preferentially over O_2 evolution due to high **oxygen overpotential** on Pt.
  - **Aqueous** $CuSO_4$ **(Pt electrodes):** Cathode: $Cu^{2+} + 2e^- \to Cu(s)$; Anode: $2H_2O \to O_2(g) + 4H^+ + 4e^-$.
  - **Aqueous** $CuSO_4$ **(Cu electrodes):** Cathode: Cu deposition; Anode: Cu dissolution ($Cu \to Cu^{2+} + 2e^-$); concentration remains constant.

7.6 Commercial Batteries & Mechanism of Corrosion

Batteries & Corrosion Reaction Architectures

- **Lead-Acid Storage Battery (Secondary Reversible Cell):**
  - **Discharging Half-Reactions:**

$$\text{Anode: } Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2e^- \tag{127}$$

$$\text{Cathode: } PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(l) \tag{128}$$

$$\text{Net Discharge: } Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \longrightarrow 2PbSO_4(s) + 2H_2O(l) \tag{129}$$

Density of H_2SO_4 drops from $1.30\,g\,cm^{-3}$ to $< 1.20\,g\,cm^{-3}$. Recharging reverses all signs.
- **Hydrogen-Oxygen Fuel Cell:**

$$\text{Anode: } 2H_2 + 4OH^- \to 4H_2O + 4e^-, \qquad \text{Cathode: } O_2 + 2H_2O + 4e^- \to 4OH^- \tag{130}$$

Theoretical thermodynamic efficiency: $\eta = \frac{\Delta G^\circ}{\Delta H^\circ} \times 100\% \approx 70\% - 75\%$ (far exceeds thermal engines).
- **Electrochemical Mechanism of Rusting of Iron:**

$$\text{Anodic site: } Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^- \quad (E^\circ = -0.44\,V) \tag{131}$$

$$\text{Cathodic site: } O_2(g) + 4H^+(aq) + 4e^- \longrightarrow 2H_2O(l) \quad (E^\circ = +1.23\,V) \tag{132}$$

$$\text{Rust Formation: } 4Fe^{2+} + O_2 + 4H_2O + 2xH_2O \longrightarrow 2Fe_2O_3 \cdot xH_2O(s) + 8H^+ \tag{133}$$

Prevented by galvanization (sacrificial zinc coating: $E^\circ_{Zn^{2+}/Zn} = -0.76\,V < E^\circ_{Fe^{2+}/Fe} = -0.44\,V$).

8 Chemical Kinetics (Unit 8)

8.1 Reaction Rate, Rate Laws & Order vs Molecularity

For a general reaction:

$$aA + bB \longrightarrow cC + dD \tag{134}$$

The instantaneous stoichiometric rate of reaction is:

$$r = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt} \tag{135}$$

Experimental Differential Rate Law:

$$r = k[A]^x[B]^y \tag{136}$$

where k is the specific rate constant, x is the order with respect to A, y is the order with respect to B, and $n = x + y$ is the overall reaction order.

Feature / Property	Order of Reaction	Molecularity of Reaction
Definition	Sum of powers of reactant concentrations in experimentally determined rate law.	Number of reacting species (atoms, ions, molecules) colliding simultaneously in an elementary step.
Determination	Strictly experimental parameter; cannot be deduced from balanced equation.	Theoretical property deduced from proposed elementary reaction mechanism.
Allowed Values	Can be zero, positive, negative, or fractional $(0, 1, 2, 3, 1/2, -1)$.	Always a positive non-zero integer $(1, 2, 3)$. Cannot be zero or fractional!
Upper Bound	Can exceed 3.	Rarely exceeds 3 due to vanishing statistical probability of ternary simultaneous collisions.
Complex Reactions	Applicable to both elementary and overall complex reactions.	Meaningless for a complex multistep reaction; applies strictly to individual elementary steps.

Table 6: Systematic Comparison: Order of Reaction vs Molecularity.

Units of Rate Constant (k):

$$\text{Units of } k = \left(\text{mol L}^{-1}\right)^{1-n}\text{s}^{-1} = (\text{atm})^{1-n}\,\text{s}^{-1} \tag{137}$$

- Zero-order ($n = 0$): $\text{mol L}^{-1}\,\text{s}^{-1}$.
- First-order ($n = 1$): s^{-1} (completely independent of concentration units!).
- Second-order ($n = 2$): $\text{L mol}^{-1}\,\text{s}^{-1}$.

8.2 Integrated Rate Equations: Full Calculus Derivations

8.2.1 1. Zero-Order Kinetics

For $A \rightarrow$ Products, rate is independent of $[A]$:

$$-\frac{d[A]}{dt} = k[A]^0 = k \implies d[A] = -k\,dt \tag{138}$$

Integrating from $t = 0$ ($[A] = [A]_0$) to time t ($[A] = [A]_t$):

$$\int_{[A]_0}^{[A]_t} d[A] = -k\int_0^t dt \implies [A]_t - [A]_0 = -kt \implies [A]_t = [A]_0 - kt \tag{139}$$

Half-Life ($t_{1/2}$) and Completion Time ($t_{100\%}$): At $t = t_{1/2}$, $[A]_t = \frac{[A]_0}{2}$:

$$\frac{[A]_0}{2} = [A]_0 - kt_{1/2} \implies t_{1/2} = \frac{[A]_0}{2k} \quad \text{(Half-life is directly proportional to initial concentration!)} \tag{140}$$

At $t = t_{100\%}$, $[A]_t = 0$:

$$0 = [A]_0 - kt_{100\%} \implies t_{100\%} = \frac{[A]_0}{k} = 2t_{1/2} \tag{141}$$

Zero-order reactions proceed to complete termination in a finite time. **Examples:** Photochemical synthesis of HCl over water ($H_2 + Cl_2 \xrightarrow{h\nu} 2HCl$); decomposition of NH_3 on hot Pt/W surface at high pressure.

8.2.2 2. First-Order Kinetics

For $A \to$ Products:

$$-\frac{d[A]}{dt} = k[A] \implies \frac{d[A]}{[A]} = -kdt \tag{142}$$

Integrating from $t = 0$ ($[A] = [A]_0$) to time t ($[A] = [A]_t$):

$$\int_{[A]_0}^{[A]_t} \frac{d[A]}{[A]} = -k\int_0^t dt \implies \ln\left(\frac{[A]_t}{[A]_0}\right) = -kt \implies [A]_t = [A]_0 e^{-kt} \tag{143}$$

$$k = \frac{1}{t}\ln\left(\frac{[A]_0}{[A]_t}\right) = \frac{2.303}{t}\log_{10}\left(\frac{[A]_0}{[A]_t}\right) = \frac{2.303}{t}\log_{10}\left(\frac{a}{a-x}\right) \tag{144}$$

Half-Life ($t_{1/2}$): At $t = t_{1/2}$, $[A]_t = \frac{[A]_0}{2}$:

$$k = \frac{2.303}{t_{1/2}}\log_{10}(2) \implies t_{1/2} = \frac{\ln 2}{k} = \frac{0.69315}{k} \approx \frac{0.693}{k} \tag{145}$$

The half-life of a first-order reaction is **strictly independent of initial reactant concentration**.

NCERT Paradigm: Fractional Lifetimes of First-Order Reactions

For any first-order process ($[A]_t = [A]_0 e^{-kt}$), each half-life interval reduces the remaining reactant concentration by exactly 50%, regardless of initial amount. Rapid shortcuts for NEET:

$$t_{75\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{25}\right) = \frac{2.303}{k}\log_{10}(4) = 2 \times \frac{0.693}{k} = 2t_{1/2}$$

$$t_{87.5\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{12.5}\right) = \frac{2.303}{k}\log_{10}(8) = 3t_{1/2}$$

$$t_{93.75\%} = 4t_{1/2} \quad \text{(remaining fraction } = 6.25\%)$$

$$t_{99\%} = \frac{2.303}{k}\log_{10}(100) = \frac{2 \times 2.303}{k} \approx \frac{4.606}{k} \approx 6.64t_{1/2}$$

$$t_{99.9\%} = \frac{2.303}{k}\log_{10}\left(\frac{100}{0.1}\right) = \frac{3 \times 2.303}{k} \approx \frac{6.909}{k} \approx 10t_{1/2}$$

NCERT Benchmark Conclusion: The time required for 99.9% completion of a first-order reaction is approximately 10 times its half-life ($t_{99.9\%} \approx 10t_{1/2}$), while $t_{99\%} \approx 2t_{90\%}$.

Gas Phase First-Order Decomposition ($A(g) \to B(g) + C(g)$): If initial pressure of A is P_i and total pressure at time t is P_t:

$$P_t = (P_i - p) + p + p = P_i + p \implies p = P_t - P_i \implies P_A = P_i - p = 2P_i - P_t \tag{146}$$

$$k = \frac{2.303}{t}\log_{10}\left(\frac{P_i}{2P_i - P_t}\right) \tag{147}$$

8.2.3 3. Pseudo-First-Order Kinetics

Reactions of higher order that behave kinetically as first-order when one reactant is present in large stoichiometric excess:

- **Acid-Catalyzed Ester Hydrolysis:**

$$\mathrm{CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH} \tag{148}$$

Because water is solvent ($[\mathrm{H_2O}] \approx 55.5\,\mathrm{M} \approx$ constant):

$$r = k'[\text{ester}][\mathrm{H_2O}] = k[\text{ester}], \qquad k = k'[\mathrm{H_2O}] \tag{149}$$

Tracked by titrating aliquots with standard NaOH: $k = \frac{2.303}{t}\log_{10}\left(\frac{V_\infty - V_0}{V_\infty - V_t}\right)$.

- **Inversion of Cane Sugar (Polarimetry):**

$$\mathrm{C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6\ (\text{D-glucose}) + C_6H_{12}O_6\ (\text{D-fructose})} \tag{150}$$

The optical rotation changes from dextro ($+66.5°$) to levo ($-39.7°$). Kinetics tracked via polarimeter angle of rotation:

$$k = \frac{2.303}{t}\log_{10}\left(\frac{r_0 - r_\infty}{r_t - r_\infty}\right) \tag{151}$$

8.3 Temperature Dependence: The Arrhenius Equation

Svante Arrhenius (1889) derived the exponential relationship between reaction rate constant and absolute temperature:

$$k = Ae^{-E_a/RT} \tag{152}$$

where A is the pre-exponential factor (frequency factor), E_a is the activation energy, and $e^{-E_a/RT}$ represents the Boltzmann fraction of molecular collisions having kinetic energy $\geq E_a$.

$$\ln k = \ln A - \frac{E_a}{RT} \iff \log_{10} k = \log_{10} A - \frac{E_a}{2.303RT} \tag{153}$$

An Arrhenius plot of $\log_{10} k$ versus $\frac{1}{T}$ produces a straight line:

$$\text{Slope} = -\frac{E_a}{2.303R}, \qquad y\text{-intercept} = \log_{10} A \tag{154}$$

Two-Temperature Comparative Formulation:

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right] = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] \tag{155}$$

8.4 Collision Theory & Catalysis Kinetics

According to Max Trautz and William Lewis:

$$k = P \cdot Z_{AB} \cdot e^{-E_a/RT} \tag{156}$$

where Z_{AB} is the collision frequency ($10^{25} - 10^{28}\,\text{collisions}\,\text{s}^{-1}\,\text{cm}^{-3}$), and P is the steric or orientation factor representing the fraction of collisions occurring with proper spatial orientation.

Catalysis Kinetics & Activation Energy Reduction

A catalyst accelerates reaction rate by providing an alternative reaction pathway involving an activated complex with a substantially lower activation energy E_a' ($E_a' < E_a$):

$$\frac{k_{\text{cat}}}{k_{\text{uncat}}} = \frac{Ae^{-E_a'/RT}}{Ae^{-E_a/RT}} = e^{(E_a - E_a')/RT} = e^{\Delta E_a/RT} \tag{157}$$

Critical Examination Invariants:

- A catalyst reduces activation energy for **both forward and backward reactions by the identical amount**: $\Delta E_{a,f} = \Delta E_{a,b}$.
- A catalyst does **NOT** alter the overall enthalpy of reaction: $\Delta_r H = E_{a,f} - E_{a,b} =$ constant.
- A catalyst does **NOT** alter the equilibrium constant K_{eq} or equilibrium composition.

- A catalyst does **NOT** alter Gibbs free energy $\Delta_r G$ or reaction spontaneity.